

Multiple Choice

1. **Answer:** B

Options: A) Water; B) Starches; C) Sugars; D) Amino acids; E) Proteins

Note: Saliva contains the enzyme amylase, which specifically catalyzes the breakdown of starches into simpler sugars during the digestive process.

2. **Answer:** A

Options: A) Alters the host's immune response to cancer cells; B) Mutates the host's DNA, leading to uncontrolled cell growth; C) Stimulates cell division, leading to overgrowth and cancer formation; D) Acts as a co factor for a cellular oncogene; E) Produces toxins that damage cellular DNA

Note: The papilloma family of viruses, particularly human papillomavirus (HPV), can cause cancer by evading the host's immune response, allowing infected cells to proliferate unchecked and leading to tumor development.

3. **Answer:** E

Options: A) Herbal remedies; B) Use of antibiotics; C) Regular intake of vitamins; D) Administration of tetanus vaccine; E) Attention to sewage control and hygiene

Note: Attention to sewage control and hygiene is essential for polio eradication because the virus is transmitted through the fecal-oral route, and improving sanitation and hygiene practices significantly reduces transmission risks.

4. **Answer:** A

Options: A) Type IIX fibres.; B) Type III fibres.; C) Type VII fibres.; D) Type V fibres.; E) Type Ia fibres.

Note: Type IIX fibers, also known as fast-twitch glycolytic fibers, fatigue quickly during all-out exercise like sprinting due to their reliance on anaerobic metabolism, which leads to rapid accumulation of lactic acid and depletion of energy substrates.

5. **Answer:** E

Options: A) recurrent laryngeal nerve which is the afferent limb of the cough reflex.; B) internal laryngeal nerve which is the afferent limb of the gag reflex.; C) vagus nerve which is the afferent limb of the gag reflex.; D) vagus nerve which is the afferent limb of the cough reflex.; E) internal laryngeal nerve which is the afferent limb of the cough reflex.

Note: The internal laryngeal nerve is responsible for providing sensory innervation to the mucosa of the upper larynx and plays a crucial role as the afferent limb in the cough reflex, making it the correct answer.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: Collagen fibres are essential for tissue repair, but their synthesis and maturation require a complex cascade of cellular and molecular events, leading to a time-dependent re-establishment during the wound healing process.

7. **Answer:** True

Options: A) True; B) False

Note: Rising incomes and urbanization typically lead to dietary shifts towards higher consumption of meat and dairy, as these products are often associated with increased socioeconomic status and urban lifestyles.

8. **Answer:** False

Options: A) True; B) False

Note: Transamination is a reversible reaction that occurs in various tissues throughout the body, not just in the liver, as it plays a crucial role in amino acid metabolism.

9. **Answer:** True

Options: A) True; B) False

Note: A deficiency of vitamin D impairs calcium absorption and bone metabolism, which can lead to enamel defects and an increased susceptibility to dental caries.

10. **Answer:** False

Options: A) True; B) False

Note: The pulse is felt during ventricular contraction, not atrial contraction, because it is the forceful ejection of blood from the ventricles into the arteries that creates the palpable pulse.

Long Form Response

11. **Answer:** An increase in survival time for individuals with a disease, while keeping all other factors constant, will lead to a rise in the prevalence of that disease within the population. This is because longer survival times allow individuals to remain in the population for extended periods, thereby maintaining or increasing the number of active cases. Additionally, as more individuals live with the disease longer, the cumulative incidence within the population rises, further elevating prevalence rates. Consequently, health systems may face increased demands for management and care of these chronic cases, highlighting the importance of understanding the dynamics between survival and prevalence in disease epidemiology.

Note: correct because increased survival time allows individuals with the disease to remain in the population longer, thereby increasing the number of active cases and contributing to higher prevalence rates.

12. **Answer:** The relationship between dietary calcium intake and absorption efficiency is characterized by an inverse correlation; as calcium intake increases, the body's efficiency in absorbing calcium tends to decrease. When calcium levels are low, the intestine enhances its absorption mechanisms, allowing for a greater proportion of dietary calcium to be utilized. Conversely, at higher intake levels, the body may become less efficient due to factors such as saturation of transport systems and feedback inhibition. This dynamic suggests that optimal calcium absorption occurs at lower intake levels, where the body can maximize its utilization of available calcium. Understanding this relationship is crucial for designing dietary recommendations that ensure adequate calcium levels while promoting efficient absorption.

Note: correct because it accurately describes the inverse relationship between dietary calcium intake and absorption efficiency, noting that lower intake stimulates enhanced absorption mechanisms while higher intake can lead to saturation and reduced efficacy in absorption.